Natural Language Processing Course

Project Description

Milestone 1: Data Preprocessing Pipeline

Milestone Description

The objective of **Milestone 1** is to establish the foundation of your NLP project by defining the problem, selecting and analyzing an appropriate dataset, and implementing an initial NLP preprocessing pipeline.

By the end of this milestone, each team must demonstrate a clear understanding of the selected problem, the characteristics of its data, and the preprocessing steps required to prepare the data for subsequent NLP modeling.

Milestone Deliverables:

The project must satisfy the following requirements:

1. **Problem Definition:** Submit a brief description containing:

- Project title.
- Problem statement and motivation.
- At least 3 Project objectives.
- Target users or application domain.
- At least one research question.
- Description of the NLP task to be implemented.

2. **Dataset**: Submit the direct dataset link or a link to the repository/source from which the dataset can be obtained. The submission must include:

- Dataset name and sources.
- Direct URL or repository link.
- Number of records or documents.
- Description of the available fields and target labels, where applicable.
- Language(s) used in the dataset.
- Data collection methods and source.

3. **Exploratory Data Analysis:** Provide at least 4 visualizations or statistical analyses covering the following:

- Dataset overview: number of records, fields, and relevant attributes.
- Data quality: missing values, duplicate records, empty texts, and other relevant quality issues.

- Text characteristics: text length, document length, or token distribution, as appropriate.
- Task-specific analysis: investigate characteristics that may affect the selected NLP task.

Examples of task-specific insights:

1. **Text classification:** class distribution, class imbalance, and representative examples from each class.
2. **Sentiment analysis:** sentiment distribution and differences between positive, negative, and neutral texts.
3. **Fake news detection:** distribution of real and fake news, article length, and potential duplicate or misleading records.
4. **Question answering or RAG:** number of documents, document length, topic coverage, and availability of relevant questions or answers.
5. **Text summarization:** document and summary lengths, summary coverage, and availability of reference summaries.
6. **Arabic NLP:** language and dialect variation, spelling inconsistencies, code-switching, and other relevant linguistic characteristics.

4. **NLP Pipeline:** Implement an initial, executable preprocessing pipeline using Python and appropriate NLP libraries. The pipeline must include at least four applicable preprocessing operations, selected according to the nature of the dataset and the project objectives.

The implementation must demonstrate the pipeline using at least five representative input examples, showing the original text and its processed output. Students must explain the purpose of each selected operation and justify why it is appropriate for their dataset. Preprocessing operations that may remove useful information should not be applied automatically.

Required output: A runnable Jupyter Notebook or Google Colab notebook containing the preprocessing implementation and example outputs.

Submission Guidelines:

1. Submit all deliverables through the designated course platform by the 6th of November at 12:00p.m.by only **ONE** member from the team.
2. Ensure that the dataset link works and that the notebook can be **executed**.
3. Include appropriate references for datasets and external resources.
4. Clearly identify the **contributions** of each team member.
5. The dataset analysis and preprocessing pipeline must be relevant to the selected project.
6. Teams are expected to use the results of this milestone as the foundation for model development and experimentation in Milestone 2.